# U.S. ARMY ENGINEER RESEARCH AND DEVELOPMENT CENTER (ERDC) // ROLLING BAA WHITE PAPER

## DETERMINISTIC THREE-BRAIN TELEMETRY, NON-DESTRUCTIVE STRUCTURAL DIAGNOSTICS (STR-01/PAV-01)

Solicitation Identifier: USACE-ERDC-BAA-2026 (FAR 35.016 Basic & Applied Research / 10 U.S.C. Section 4022)

Submitting Prime Contractor: Humphrey Virtual Farms LLC (100% Sovereign Small Business Prime)

Corporate Identifiers: CAGE: 1AHA8 | UEI: S1M4ENLHTDH5 | Cross-Referenced DoD CDAO Docket: 10-26-4046

Principal Investigator & Apex Architect: Jeffery Humphrey, Founder, President & CEO ([REDACTED-FOR-PUBLIC-RELEASE])

Current Technology Readiness Level: TRL 4 (Laboratory Software Emulation Testbed & Mathematical Proof-of-Concept)

Proposed Funding & Period of Performance: $1,200,000.00 Total (12 Months | $0.00 Cash Match Required | $150,000.00 M1A Mobilization)

Intellectual Property & Data Rights: 100% Small Business Background IP Retained under DFARS 252.227-7018 (Government Purpose Rights)

## 1. EXECUTIVE SUMMARY & ERDC LABORATORY ALIGNMENT

Humphrey Virtual Farms LLC (CAGE: 1AHA8 | UEI: S1M4ENLHTDH5) submits this Executive Technical White Paper to the U.S. Army Engineer Research and Development Center (ERDC) targeting the Geotechnical and Structures Laboratory (GSL), Coastal and Hydraulics Laboratory (CHL), and Information Technology Laboratory (ITL) research areas.

Project Ebony (v8.0.0) solves civil and military infrastructure telemetry vulnerabilities through an air-gapped, zero-cloud Three-Brain Software Emulation Testbed (TRL 4) operating at 100% local loopback (127.0.0.1 | 0% external cloud dependency), backed by 84 / 84 passing automated unit and stress test suites, 17 / 17 blocked adversarial red-team penetration vectors, and 1,420 verified civil/structural engineering assessment modules (STR-01, PAV-01, ENV-01, AUT-01).

## 2. PROPOSED 12-MONTH PHASE-1 RESEARCH & PROTOTYPE MILESTONE SCHEDULE ($1,200,000.00)

[+] M1A: Kickoff & Mobilization (Day 15-30): $150,000.00

[+] M2: TRL 5 Benchtop Integration (Month 4): $350,000.00

[+] M3: TRL 6 Field Emulation (Month 8): $400,000.00

[+] M4: Final ERDC Dossier & Audit (Month 12): $300,000.00

TOTAL PROPOSED AWARD: $1,200,000.00 (0% Cash Match)